

JEREMIAH DANIEL SERENGE

Data Engineer | MLOps Specialist | Business Intelligence Developer

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PROFESSIONAL SUMMARY

Data Engineer with 6+ years specializing in financial services analytics, credit risk modeling, and business intelligence solutions. Proven track record building production-grade ML systems for fintech organizations with measurable business impact: 16.3% improvement in credit default prediction accuracy, 86% automation of customer queries, and 20+ hours weekly savings through Power BI dashboard implementation. Expert in Python, machine learning, statistical analysis, Azure cloud infrastructure, and translating complex data insights into actionable business strategies for stakeholder decision-making.

PROFESSIONAL EXPERIENCE

Data Engineer | BI Developer

Dataposit Africa, Nairobi, Kenya | Jan 2024 – Present

- Designed and maintained scalable ETL pipelines integrating ERP platforms (Microsoft Dynamics 365 Business Central, SAP, and other enterprise systems) into analytics-ready data models using Azure cloud services and Microsoft Fabric, supporting enterprise BI reporting and downstream machine learning workloads.
- Built and optimised Python-based data engineering pipelines on Azure Databricks and Apache Spark to process high-volume OLTP and operational datasets, improving data quality, processing efficiency, and scalability across multiple client environments.
- Developed end-to-end Microsoft Power Platform solutions using Power Apps and Power Automate to digitise business processes, automate approvals and workflows, capture operational data, and integrate seamlessly with Power BI for real-time reporting and decision support.
- Designed automated workflows with Microsoft Power Automate to streamline finance, HR, operations, and customer service processes, reducing manual intervention and improving operational efficiency.
- Engineered Microsoft Copilot agents that automate business workflows, including intelligent helpdesk triage, HR self-service, knowledge retrieval, and sales support, significantly reducing response times and enhancing employee productivity.
- Integrated Microsoft Copilot with Power BI semantic models to enable natural language analytics, allowing business users to query enterprise data conversationally and generate actionable insights without requiring technical expertise.
- Integrated Model Context Protocol (MCP) services into analytics workflows, enabling AI agents to access enterprise datasets directly, automate complex data analysis tasks, and eliminate reliance on manual spreadsheet processing and traditional Excel-based analysis.
- Built end-to-end machine learning pipelines using Azure AI Studio and the Azure AI ecosystem, covering data preparation, feature engineering, model training, validation, deployment, and monitoring through Python SDKs and FastAPI services.
- Developed AI-powered forecasting and predictive analytics solutions using Azure AI Studio's managed models and custom machine learning approaches to support demand forecasting, inventory optimisation, supply chain planning, and cost optimisation across business operations.
- Collaborated closely with Finance, HR, Operations, and executive stakeholders to translate complex business requirements into scalable data architectures, AI solutions, enterprise applications, and interactive BI reporting platforms that support strategic decision-making.

Freelance Data Scientist & Analytics Consultant

Upwork, Fiverr, & Direct Client Engagements | Jan 2019 – Dec 2023

Credit Risk Modeling & Portfolio Analytics

- Built credit scoring models using Logistic Regression and Random Forest, achieving 96% classification accuracy on loan default prediction for Singaporean microfinance client.
- Engineered features from repayment history, transaction behavior, and demographic data to improve model signal quality and reduce false negatives, achieving 16.3% improvement in default prediction accuracy.

- Designed and developed new credit products informed by model outputs, segmenting customers by risk profile to match loan terms with creditworthiness.
- Built loan recovery analytics framework to identify high-risk delinquent accounts, prioritize recovery actions, and track recovery performance over time.
- Conducted root cause analysis on default clusters, identifying key behavioral and economic drivers; model directly contributed to 15% reduction in default rates in first quarter post-deployment.
- Finance Company Analytics Pipeline Optimization (Nairobi-based)
- Partnered with multiple Nairobi-based finance companies to redesign and automate analytical pipelines, reducing manual reporting turnaround time by 60% and improving data quality through systematic validation.
- Implemented robust ETL processes connecting legacy finance systems to modern analytics platforms, enabling real-time portfolio and performance reporting.

Statistical Analysis & Business Intelligence

- Performed complex statistical analyses using STATA and SPSS for international finance and academic clients, including regression, panel data, and survival analysis.
- Built predictive models and visualization dashboards translating complex datasets into clear, decision-ready business insights.
- Conducted data mining to discover behavioral patterns from large financial datasets informing strategic decisions.

PROJECTS & ACHIEVEMENTS

- **Production-ready Credit Scoring Model:** Developed logistic regression & Random Forest-based credit model for Singaporean finance firm with 99% uptime and achieved 16.3% improvement in default prediction accuracy.
- **MoPhones Portfolio Case Study:** Conducted comprehensive portfolio analysis on 109,702 loan accounts using Jupyter and Streamlit with Plotly visualizations, demonstrating advanced EDA and portfolio profiling techniques. Available on Streamlit: <https://share.streamlit.io/user/jsereinge>
- **Predictive Sales Forecasting:** Forecasted retail sales using SARIMAX and reinforcement learning on Azure Databricks.
- **AI-Powered Chatbot for Customer Support:** Designed customer-facing chatbot with FastAPI and Gemini LLM for NLP-based service automation, integrated with MongoDB. Which achieved 86% customer query resolution without human intervention.
- **Big Data Pipeline Design:** Engineered scalable and efficient big data pipelines using Hadoop and Apache Spark, enabling high-throughput analytics and seamless data ingestion for Azure-hosted ML models via Azure SDKs. Optimized for performance, reliability, and integration with enterprise AI workflows.
- **Enterprise Business Intelligence Solutions:** Designed and deployed enterprise-grade Power BI dashboards for multiple Kenyan organizations, enabling real-time decision-making and eliminating manual Excel-based reporting. Implemented automatic and incremental data refresh mechanisms to optimize system performance, reduce latency, and support seamless user experiences across departments.
- **Financial Data ETL Processes:** Developed robust SQL-based and Python ETL processes integrating financial data from multiple ERP sources (Business Central, SAP), ensuring data quality and consistency for model training and compliance reporting.

CORE COMPETENCIES

1. **Data Engineering & ETL/ELT** – Designing and orchestrating scalable data pipelines using Azure, Python, Apache Spark, Databricks, SQL, and DBT, with ERP integrations (Microsoft Dynamics 365 Business Central, SAP).
2. **Advanced Data Analytics** – Performing statistical analysis, exploratory data analysis (EDA), forecasting, anomaly detection, financial modelling, and business insights using Python (pandas, NumPy, scikit-learn) and Microsoft Excel.

3. **Machine Learning & Predictive Analytics** – Developing end-to-end ML solutions for credit risk modelling, demand forecasting, inventory optimisation, customer segmentation, recommendation systems, and predictive modelling using Logistic Regression, Random Forest, XGBoost, and other ensemble techniques.
4. **Azure AI & MLOps** – Building production-ready AI solutions using Azure AI Studio, Azure Machine Learning, FastAPI, REST APIs, model deployment, monitoring, versioning, and CI/CD practices.
5. **Enterprise AI & Copilot Solutions** – Developing Microsoft Copilot agents, integrating LLM-powered workflows, Model Context Protocol (MCP) servers, and conversational AI solutions for enterprise knowledge retrieval and business automation.
6. **Power Platform Development** – Building end-to-end business applications using Power Apps and Power Automate, automating workflows and integrating seamlessly with Power BI for operational reporting and process optimisation.
7. **Business Intelligence & Data Modelling** – Designing dimensional data models, Power BI semantic models, DAX measures, executive dashboards, KPI reporting, and self-service analytics solutions.
8. **Cloud Data Platforms & Software Engineering** – Developing scalable cloud-native solutions using Azure, Databricks, Git, containerisation, REST APIs, Agile methodologies, and modern software engineering best practices.
9. **Financial Analytics & Risk Modelling** – Delivering analytics solutions for credit scoring, portfolio analytics, repayment behaviour analysis, probability of default (PD), Expected Credit Loss (ECL), underwriting, and financial forecasting.
10. **Stakeholder Engagement & Solution Delivery** – Partnering with cross-functional teams across Finance, HR, Operations, Procurement, and Supply Chain to translate business requirements into scalable data, AI, and analytics solutions that drive strategic decision-making.

EDUCATION & CERTIFICATIONS

BSc Applied Statistics with Computing

Relevant coursework: Statistical Modelling, Probability Theory, Regression Analysis, Data Structures, Database Systems, Operations Research

Professional Certifications:

- Power BI Data Analyst — PwC Switzerland Virtual Experience
- Machine Learning with Python — IBM
- Data Science Tools & Methodology — IBM
- AI Skills Fest 2026 – Microsoft
- Fundamentals of Quantitative Modeling – Wharton University of Pennsylvania